# Deepak Sathyan

*Curriculum Vitae*

arXiv | InspireHEP | Google Scholar | ORCID | Website

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Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University, College Station, TX

## Experience

**Postdoctoral Research Associate** — College Station, TX
Mitchell Institute for Fundamental Physics and Astronomy — Sept 2024 – Aug 2027
Texas A&M University — 3 years

## Education

**University of Maryland** — College Park, MD
**Ph.D. in Physics** — Aug 2018 – July 2024
Thesis: Unifying Searches for New Physics with Precision Measurements of the W Boson Mass
Advisor: Dr. Kaustubh Agashe

**Boston University** — Boston, MA
**Bachelor of Arts in Physics, Summa Cum Laude with Honors** — Aug 2014 – May 2018
GPA: 3.9
Minor: Mathematics
Thesis: Muon (g-2): Searching for the Muon Electric Dipole Moment
Advisor: Dr. Robert Carey

## Research Interests

**Beyond the Standard Model searches with Collider Physics: LHC and muon collider phenomenology, Randall-Sundrum and baryon-number-violation models**

**Searches for sub-GeV Dark Matter: dark matter-neutrino interactions, dark photon and dark sector production, cosmic ray-dark matter scattering**

**Neutrino Physics: supernova neutrino attenuation and detection, open-system observables, neutrinophilic mediators, neutrino NSI**

## Seminars

**9. HEP Seminar at Northwestern University** — Sept 2026
Cosmic Ray–Dark Matter scattering explanation of the Galactic Center Excess

**8. Fermilab Cosmic Physics Center Seminar** — Sept 2026
Cosmic Ray–Dark Matter scattering explanation of the Galactic Center Excess

**7. CMS Exotica general meeting** May 2024
Unification of Searches and Measurements: Probing BSM with the W boson mass measurement

**6. Seminar at Texas A&M** Dec 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

**5. Seminar at UT Austin** Dec 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

**4. CMS SUSY general meeting** Nov 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

**3. HEP Seminar at University of Pittsburgh** Nov 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

**2. HEP Seminar at Northwestern University** Oct 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

**1. Theory Seminar at Washington University in St. Louis** Oct 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

## Conference Talks

**14. Light Dark World 2026 at Carleton University [Invited]** July 2026
Can the LHC be sensitive to the Light Dark World?

13. CETUP June 2026
*Workshop [Invited]**
Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering

**12. Particle Physics on the Plains at the University of Kansas [Contrib.]** Nov 2025
A Baryon and Lepton Number Violation Model Testable at the LHC

11. CETUP June 2025
*Workshop [Invited]**
Can the LHC be sensitive to light dark mediators?

**10. Phenomenology Conference at the University of Pittsburgh [Contrib.]** May 2025
Can the LHC be sensitive to light dark mediators?

**9. Particle Physics on the Plains at the University of Kansas [Contrib.]** Nov 2024
A comprehensive analysis of supernova neutrino-dark matter interactions

**8. Texas TACOS @ UT Austin [Invited]** Oct 2024
A comprehensive analysis of supernova neutrino-dark matter interactions

**7. NuFact 2024 Conference at Argonne National Laboratory [Invited]** Sept 2024
A comprehensive analysis of supernova neutrino-dark matter interactions

**6. Mitchell Conference at Texas A&M University [Invited]** May 2024
A comprehensive analysis of supernova neutrino-dark matter interactions

**5. Particle Physics on the Plains at the University of Kansas [Contrib.]** Oct 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

**4. Phenomenology Conference at the University of Pittsburgh [Contrib.]** May 2023
Probing Dark Matter-Neutrino Interactions via Supernova Neutrinos

**3. Phenomenology Conference at the University of Pittsburgh [Contrib.]** May 2022
Model-Independent Measurement of Top Quark Mass Using B-Hadron Decay Lengths (Part I)

**2. April APS Meeting [Contrib.]** Apr 2021
Signals of KK graviton from extended warped extra dimensions at the LHC (II)

**1. Phenomenology Conference at the University of Pittsburgh [Contrib.]** May 2020
Signals of KK graviton from extended warped extra dimensions at the LHC (II)

## Teaching

**Principles of Modern Physics** Spring 2024
Teaching Assistant, University of Maryland

**Electricity and Magnetism I** Spring 2024
Teaching Assistant, University of Maryland

**Mathematical Methods for Physics II** Fall 2021, Fall 2022
Teaching Assistant, University of Maryland

**Fundamentals of Physics I** Summer 2019, Spring 2022
Teaching Assistant, University of Maryland

**Introduction to Quantum Mechanics I (Graduate)** Fall 2020
Teaching Assistant, University of Maryland

**Fundamentals of Physics for Biologists II** Spring 2019, Spring 2020
Teaching Assistant, University of Maryland

**Fundamentals of Physics for Biologists** Fall 2018, Fall 2019
Teaching Assistant, University of Maryland

**Guest Lecture for Undergraduate E&M at University of Maryland** Mar 2024
Lectured on vector potentials in E&M

**Guest Lecture for Graduate E&M at Texas A&M University** Oct 2025
Lectured on relativistic Larmor formula and charged particle motion in magnetic fields

## Service

**The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2026**
Conference Organizer — Texas A&M University, May 27-30, 2026

**The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2025**
Conference Organizer — Texas A&M University, May 13-16, 2025

**Mitchell Institute High Energy Phenomenology, Experiment, and Cosmology Seminar Series**
Weekly Seminar Organizer - Texas A&M University, Fall 2024 - Present

## Awards

**Breakthrough Prize in Fundamental Physics** Apr 2026
Awarded to the Muon g-2 Collaboration; recognized as a collaboration member

**Ralph Myers & Friends of Physics Award** Dec 2019
Awarded by the University of Maryland Department of Physics to outstanding Graduate Teaching Assistants

**Ralph Myers & Friends of Physics Award** May 2020
Awarded by the University of Maryland Department of Physics to outstanding Graduate Teaching Assistants

**Alumni Award** May 2018
Awarded by the Boston University Department of Physics

## Schools Attended

**GGI Lectures on the Theory of Fundamental Interactions** Jan 2023

**TASI 2024: The Frontiers of Particle Theory** June 2024

## Outreach

**BU Student Mentorship Program PRISM** 2016 - 2018
Mentored incoming freshman undergraduates in physics, providing guidance on courses, research opportunities, and resources

**BU Outreach for Local High Schools** Jan 2018
Demonstrated measurement of the charge-to-mass ratio of the electron for local high schoolers in physics

**TAMU Physics and Engineering Festival** 2025 - present

Presented demonstrations of fluid dynamics to the general public, including high school students and families

**University of South Dakota talk: "My path through particle physics research so far"** Dec 2024

Presented my path through particle physics research to students, showing a variety of interesting ideas to pursue

## Skills

**Programming:**

Mathematica, Python, C++

**Physics Software:**

MadGraph5_aMC@NLO, Pythia6, Pythia8, Delphes, ROOT, GEANT4, FeynRules

## References

**Dr. Kaustubh Agashe**

University of Maryland, College Park — kagashe@umd.edu

**Dr. Bhaskar Dutta**

Texas A&M University — dutta@tamu.edu

**Dr. Bhupal Dev**

Washington University in St. Louis — bdev@wustl.edu

**Dr. Jason Kumar**

University of Hawai'i — jkumar@hawaii.edu